

Finalization of the Standard Explanation / Recommendation for VHAR

GeomHASS - JNBalboa / ACPidlaoan / ADAquino

30 May 2024

Existing VHAR Template...

- ❑ based on the existing Explanations and Recommendations used from previous HARs and existing hazard maps
- ❑ different for a specific volcano (ex. Taal, Mayon, Pinatubo, Bulusan,...etc.)
- ❑ HARs with PAV assessments (no hazard map): lifted from previously drafted letters (OGLs/HIMs).
- ❑ generic explanation/recommendation for AV/PAV assessment with no hazard map

Results of the previous consultation (March 2024):

*Use generic explanation/ recommendation applicable for all volcanoes.

If PRONE to LAVA FLOWS (Explanation)

- *Lava flows are rivers of hot molten rock that can burn and bury land and make these lands unusable for a long period of time.*

If PRONE to PDC/ BASE SURGES (Explanation)

- *Pyroclastic Density Currents (PDCs), such as flows, surges and base surges are turbulent, fast-moving hot mixtures of volcanic fragments and gas that sweep down/away from an erupting volcano. PDCs can cause loss of life due to burning, impact force, abrasion, asphyxiation, and extreme physical injuries, burial, and damage to agriculture, livestock, and properties.*

Add-on statement if PRONE to LAVA FLOWS and/or PDCs (Recommendation)

- *Avoidance is recommended for site/sites that may potentially be affected by PDCs and lava flows.*

Results of the previous consultation (March 2024):

If PRONE to VOLCANIC TSUNAMI (Explanation)

- *Volcanic tsunami can be generated by the sudden displacement of sea/lake water by the entry of large masses of volcanic material, underwater explosion, and/or rapid ground deformation during volcanic activity. Volcanic tsunamis can cause inundation and washout of communities, drowning, physical injuries, and damage to infrastructure and properties.*

Add-on statement if PRONE to VOLCANIC TSUNAMI (Recommendation)

- *Tsunami threat related to the volcanic unrests can be addressed by community preparedness and tsunami evacuation plan, with special focus on shoreline communities. ~~and prompt evacuation of communities to avoid being caught in the tsunami occurrence while evacuating. Lakeshore communities must learn to evacuate themselves when they recognize the three natural signs of tsunami, namely 1) strong ground shaking, 2) unusual rise or fall of lake level, and 3) strong or unusual sound coming from the lake.~~*

Results of the previous consultation (March 2024):

If PRONE to LAHARS (except Mayon & Pinatubo) (Explanation)

Lahars are rapidly flowing mixtures of volcanic sediment and water in rivers that drain down volcanoes, typically produced by prolonged and intense rainfall. Lahars can cause extensive burial or washout of communities that can lead to physical injuries and loss of life, damage to built environment, agriculture, livestock, infrastructure, properties, as well as long-term flooding and siltation in river systems.

MAYON and PINATUBO:

Due to the technicalities (related to existing LGU ordinances), lahar explanations/recommendations related to **Pinatubo** and **Mayon** volcanoes would still be based on the existing/ previous HARs.

Zone 1 includes areas at high susceptibility to large-magnitude lahars (more than 10 million cubic meters per event) and sediment-laden (muddy) stream flows and flashfloods.

Zone 2 includes areas at low susceptibility to large-magnitude lahars, moderate to high susceptibility to small-magnitude lahars, and high susceptibility to sediment-laden (muddy) stream flows and/or flashfloods.

Zone 3 includes areas at low susceptibility to small-magnitude lahars and high susceptibility to sediment-laden (muddy) stream flows.

Zone 4 are areas safe from lahars but susceptible to sediment-laden (muddy) stream flows or siltation.

Zone 5 are areas safe from lahars but susceptible to persistent (more than one week) or recurrent flooding and/or backflooding due to drainage blockage.

Areas highly prone to lahars are located adjacent to an active river channel that may be a potential lahar avenue in the event of an explosive eruption and additional lahar source materials become available.

Areas moderately prone to lahars are located in medial or distal portions of lahar fans, with a moderate probability of being affected by lahars and a high probability of being affected by sediment-laden streamflows.

Areas least prone to lahars are zones located in distal portions of lahar fans, or between river channels, with a low probability of being affected by lahars and by sediment-laden stream flows.

Results of the previous consultation (March 2024):

If PRONE to BALLISTIC PROJECTILES (Explanation)

- *Ballistic projectiles are fragments thrown at a close distance from the vent/crater of an erupting volcano. Ballistic projectiles can cause loss of life, physical injuries, and damage to properties due to forceful impact when hitting ground.*

ASHFALL template

- *In case of future eruptions of {AV} Volcano and other nearby volcanoes, the site may be affected by tephra fall/ ashfall depending on the **height of the eruption plume** and prevailing wind direction at the time of eruption. Generally, tephra fall/ ashfall is **thicker** near the eruption center and thins away from the volcano. **Ash can cause widespread infrastructural, agricultural, aircraft and property damage, and negative health impacts.***

- ❑ Thus, we apply the generic explanation/recommendation for **all hazards** AV/PAV assessments (except for lahars related to Mayon & Pinatubo volcanoes)



HAS-Feb-24-11367

DATE
FOR
REPRESENTED BY
PURPOSE

27 February 2024
ROGELIO V. GARCIA
JULIUS ORTIZ IRINEO
HLURB Requirement



HASS-EQ-04



VOLCANO HAZARD ASSESSMENT

LOT DESCRIPTION, LOCATION	VOLCANO DISTANCE	LAHAR	PYROCLASTIC FLOW	LAVA FLOW
		Prone	Prone	Prone

EXPLANATION AND RECOMMENDATION

- ✓ All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map provided.
- ✓ {AV} Volcano is the nearest identified active volcano to the site.
- ✓ The site is inside the ____-kilometer Permanent Danger Zone of _____ volcano
- ✓ The Permanent Danger Zone (PDZ) is a zone that can always be affected by small-scale eruptions of _____ Volcano. Human settlement is not recommended within the PDZ.

For Taal only:

- ✓ Taal Volcano Island is designated as the Permanent Danger Zone of the volcano.

AV only

> inside PDZ/DZ, ~prone hazards

> less than ~50kms or within volcano watershed

- ✓ Lava flows are rivers of hot molten rock that can damage built environments and can make lands unusable for a long period of time.
- ✓ Pyroclastic Density Currents (PDCs), such as flows, surges and base surges are turbulent, fast-moving hot mixtures of volcanic fragments and gas that sweep down/away from an erupting volcano. PDCs can cause loss of life, extreme physical injuries, burial, and damages to agricultures, livestock, and properties.
- ✓ Avoidance is recommended for sites that may potentially be affected by primary volcanic hazards, especially PDCs (pyroclastic density currents) and lava flows
- ✓ Lahars are rapidly flowing mixtures of volcanic sediments and water in rivers that drain down volcanoes, typically produced by prolonged and intense rainfall. Lahars can cause extensive burial or washout of communities, physical injuries that could lead to loss of life, damages to the natural environment, agriculture, livestock, infrastructure, properties, and long-term flooding and siltation in river systems.
- ✓ Lahar threat to people's lives can be addressed by 1) observing or implementing legal easement adjacent to river banks, as provided in existing laws, ordinances and land-use plans, and 2) community preparedness and evacuation plan. At-risk communities must learn to evacuate themselves when lahar threats are imminent.
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- ✓ This hazard assessment supersedes any previous assessment made by this office regarding the site/s.



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	+Additional column for fissure assessment	Prone	Safe	Safe

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- ✓ {AV} Volcano is the nearest identified active volcano to the site.
- ✓ The site is outside the ___-kilometer Permanent Danger Zone of _____ volcano
- ✓ The Permanent Danger Zone (PDZ) is a zone that can always be affected by small-scale eruptions of _____ Volcano. Human settlement is not recommended within the PDZ.

For Taal only (if near Batangas)

- ✓ Volcanic fissures identified are based on results from detailed field investigations. Hazard assessment may change as DOST-PHIVOLCS scientists gather more information on fissure occurrences.
- ✓ Fissures may develop further if the condition that generated them persists, as a response to geological processes. For land-use planning and disaster risk reduction purposes, at least 5 meters on both sides of the fissure or at the edge of the deformation zone may be considered as a Zone of Avoidance.
- ✓ Stakeholders are advised to report to the Institute fissures that may be observed at Taal Volcano and vicinity that need to be reflected in the fissure map.

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> outside PDZ/DZ, ~prone/~safe hazards

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+Fissures

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- ✓ This hazard assessment supersedes any previous assessment made by this office regarding the site/s.

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LOT DESCRIPTION, LOCATION	VOLCANO DISTANCE	LAHAR	PYROCLASTIC FLOW	LAVA FLOW
		Safe	Safe	Safe

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AV only

> ~safe hazards

> less than ~50kms or within volcano watershed



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- ✓ All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map provided.
- ✓ {AV} Volcano is the nearest identified active volcano to the site/s.
- ✓ Considering the distance of the site/s from the volcano, the site/s is/are **safe from volcanic hazards** such as pyroclastic density currents, lava flows, and ballistic projectiles that may originate from the volcano.
- ✓ In case of future eruptions of {AV} Volcano and other nearby volcanoes, the site may be affected by tephra fall depending on the scale of eruption and prevailing wind direction at the time of eruption. Generally, ashfall is heavier near the active vent and thins out indefinitely away from the eruption center.
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AV only

> ~safe hazards

> more than ~50kms or beyond the volcano watershed

AV with PAV HAZARD Assessment Template

For cases where the site/some sites is/are **nearer** a POTENTIALLY ACTIVE VOLCANO.

Conditions:

1. If both AV and PAV are **within 50kms** but **AV is nearer**, assess AV only.
2. If both AV and PAV are **more than 50kms** away from the site (even if PAV is nearer), assess AV only.
3. If both AV and PAV are **within 50kms** but **PAV is nearer**,
= assess both AV and PAV; may consult VOOD.

Project Name/Lot Description	Volcano Distance	Lava Flow Hazard	Pyroclastic Density Current (PDC) Hazard	Lahar Hazard
(AV/PAV Distance only if no hazard map available)	Approximately X kilometers ____ of AV			
	Approximately X kilometers ____ of PAV			
(AV/PAV with manual assessment/ or hazard maps available)	Approximately X kilometers ____ of AV	Prone/ Safe	Prone/ Safe	Prone/ Safe
	Approximately X kilometers ____ of PAV	Prone/ Safe	Prone/ Safe	Prone/ Safe
(AV/PAV Assessment if AV with hazard maps, PAV no map available)	Approximately X kilometers ____ of AV	Prone/ Safe	Prone/ Safe	Prone/ Safe
	Approximately X kilometers ____ of PAV			



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LOT DESCRIPTION, LOCATION	VOLCANO DISTANCE	LAHAR	PYROCLASTIC FLOW	LAVA FLOW
	AV Assessment	Safe	Safe	Safe
	PAV Assessment	Prone	Prone	Prone

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AV + (PAV with hazard map)

> AV hazards ~Safe

> PAV hazards ~Prone

- ✓ {PAV} Volcano is currently classified by DOST-PHIVOLCS as a potentially active volcano, which is morphologically young-looking but with no historical or analytical records of eruption.
- ✓ Lava flows are rivers of hot molten rock that can damage built environments and can make lands unusable for a long period of time.
- ✓ Pyroclastic Density Currents (PDCs), such as flows, surges and base surges are turbulent, fast-moving hot mixtures of volcanic fragments and gas that sweep down/away from an erupting volcano. PDCs can cause loss of life, extreme physical injuries, burial, and damages to agricultures, livestock, and properties.
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	PAV Assessment	Prone	Prone	Prone

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- ✓ {PAV} Volcano is currently classified by DOST-PHIVOLCS as a potentially active volcano, which is morphologically young-looking but with no historical or analytical records of eruption.



HASS-EQ-04



AV + (PAV with hazard map) just in case?

> AV hazards ~Prone

> PAV hazards ~Prone

- ✓ Lava flows are rivers of hot molten rock that can damage built environments and can make lands unusable for a long period of time.
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LOT DESCRIPTION, LOCATION	VOLCANO DISTANCE	LAHAR	PYROCLASTIC FLOW	LAVA FLOW
	AV Assessment	Safe	Safe	Safe
	PAV Assessment	Safe	Safe	Safe

EXPLANATION AND RECOMMENDATION

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HASS-EQ-04



AV + (PAV with hazard map) (~less than 50km/within watershed)
> AV hazards ~Safe
> PAV hazards ~Safe

- ✓ {PAV} Volcano is currently classified by DOST-PHIVOLCS as a potentially active volcano, which is morphologically young-looking but with no historical or analytical records of eruption.
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VOLCANO HAZARD ASSESSMENT

LOT DESCRIPTION, LOCATION	VOLCANO DISTANCE	LAHAR	PYROCLASTIC FLOW	LAVA FLOW
	AV Assessment	Safe	Safe	Safe
	PAV Assessment	Prone		

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AV + (PAV -no hazard map)

> AV hazards ~Safe

> PAV hazards ~Prone (% VOOD assess for LHR only)

- ✓ {PAV} Volcano is currently classified by DOST-PHIVOLCS as a potentially active volcano, which is morphologically young-looking but with no historical or analytical records of eruption, **and therefore, has yet to be determined its eruptive and hazard potential has yet to be determined.**
- ✓ The site is located on the alluvial fan near {PAV} and may be potentially affected by lahars in the event of severe/ prolonged precipitation which can erode and remobilize old volcanic deposits on the slopes of the volcano.
- ✓ Lahars are rapidly flowing mixtures of volcanic sediments and water in rivers that drain down volcanoes, typically produced by prolonged and intense rainfall. Lahars can cause extensive burial or washout of communities, physical injuries that could lead to loss of life, damages to the natural environment, agriculture, livestock, infrastructure, properties, and long-term flooding and siltation in river systems.
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HASS-EQ-04



AV + (PAV -no hazard map) (*rare case ie. Banahaw/ San Cristobal*)
> AV hazards ~Prone
> PAV hazards ~Safe (% VOOD assess for LHR only)

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EXPLANATION AND RECOMMENDATION

- ✓ All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map provided.
- ✓ {AV} Volcano is the nearest identified active volcano to the site.
- ✓ Considering the distance of the site/s from the volcano, the site is **safe from volcanic hazards** such as pyroclastic density currents, lava flows, and ballistic projectiles that may originate from the volcano.
- ✓ In case of future eruptions of {AV} Volcano and other nearby volcanoes, the site may be affected by tephra fall depending on the scale of eruption and prevailing wind direction at the time of eruption. Generally, ashfall is heavier near the active vent and thins out indefinitely away from the eruption center.



HASS-EQ-04



AV + (PAV -no hazard map) (~more than 50km/outside watershed)
> AV hazards ~Safe
> PAV hazards ~Safe

- ✓ {PAV} Volcano is currently classified by DOST-PHIVOLCS as a potentially active volcano, which is morphologically young-looking but with no historical or analytical records of eruption and therefore, has yet to be determined its eruptive and hazard potential has yet to be determined.
- ✓ This hazard assessment supersedes any previous assessment made by this office regarding the site/s.

Assessed by
Verified by

Raymond Patrick R. Maximo
Jayvie N. Balboa

Officer-of-the-Day
Senior Science Research Specialist

Approved by

MABELLINE T. CAHULOGAN

Supervising Science Research
Specialist

Future Discussions...

❏ FISSURES

(Taal)

- ✓ *Volcanic fissures identified are based on results from detailed field investigations. Hazard assessment may change as DOST-PHIVOLCS scientists gather more information on fissure occurrences.*
- ✓ *Fissures may develop further if the condition that generated them persists, as a response to geological processes. For land-use planning and disaster risk reduction purposes, at least 5 meters on both sides of the fissure or at the edge of the deformation zone may be considered as a Zone of Avoidance.*
- ✓ *Stakeholders are advised to report to the Institute fissures that may be observed at Taal Volcano and vicinity that need to be reflected in the fissure map.*

**(EQ) Fissure, with its related ground subsidence, may not be a potential source of earthquake. However, fissures and ground subsidence may develop further if the condition that generated them continued to persist, or as a response to strong ground shaking. The recommended buffer zone, or Zone of Avoidance, against fissuring is at least 5 meters on both sides of the fissure.*

- Maybe standardize fissure-related explanation (EQ/VOL)?:

"Furthermore, the site is within the Taal fracture zone.....? Potential fissure"

❏ LAHARS

- when to use the standard explanation/ recommendation for Mayon and Pinatubo?

STATUS OF VOLCANO-RELATED HAZARD INFORMATION / MAPS

https://docs.google.com/presentation/d/1HjjQ_6z05rc98c7ugwSP4fkiutQia-qi4ymplUaprzw/edit?usp=sharing

Future Discussions...

❑ VOLCANO HAZARD INFORMATION / MAPS UPDATE (AVs)

Active Volcanoes	HAZARD EXTENT	LAV	LHR	PDC	BAL	VTS	Remarks
Babuyan Claro	Cagayan	✓	✓	✓			
Biliran (Anas)	Biliran	✓	✓	✓			
Camiguin de Babuyanes	Cagayan	✓	✓	✓			
Didicas	Cagayan	✓	✓	✓			
Hibok-hibok	Camiguin	✓	✓	✓	✓		
Iraya	Batanes	✓	✓	✓			
Iriga	Camarines Sur			✓		✓	For revision
Smith	Cagayan	✓	✓	✓			
Taal	Batangas			✓	✓	✓	
Banahaw	Laguna, Quezon		✓	✓			For standard layout
Bud Dajo	Sulu						Ongoing
Bulusan	Sorsogon	✓	✓	✓			For standard layout
Cabalian	Southern Leyte		✓	✓			For standard layout
Cagua	Cagayan	✓	✓	✓			For revision

Active Volcanoes	HAZARD EXTENT	LAV	LHR	PDC	BAL	VTS	Remarks
Isarog	Camarines Sur	✓	✓	✓			
Kanlaon	Negros Oriental, Negros Occidental	✓	✓	✓			LHR for revision
Leonard Kniaeff	Davao del Norte, Compostela Valley		✓				
Makaturing	Lanao del Sur						
Matutum	South Cotabato, Sarangani, Sultan Kudarat		✓	✓			
Mayon	Albay	✓	✓	✓			LHR/PDC for revision
Musuan (Calayo)	Bukidnon						
Parker	South Cotabato, Sarangani		✓	✓			
Pinatubo	Bataan, Pampanga, Tarlac, Zambales		✓				
Ragang	Lanao del Sur, North Cotabato, Maguindanao	✓	✓	✓			

Ongoing?

To Verify

Not Applicable

Future Discussions...

❑ VOLCANO HAZARD INFORMATION / MAPS UPDATE (PAVs)

Potentially Active Volcanoes	HAZARD EXTENT	LAV	LHR	PDC	BAL	VTS	Remarks
Apo	Cotabato, Davao Del Sur						
Balut	Davao Occidental						
Cuernos de Negros	Negros Oriental						
Dakut	Sulu						
Gorra	Sulu						
Lapac	Sulu						
Malinao	Albay, Camarines Sur						
Malindig	Marinduque						
Mandalagan	Negros Occidental						
Maripipi	Biliran						
Mariveles	Bataan						
Natib	Bataan						Nasa ETR
Parang	Sulu						
Parangan	Sulu						

Potentially Active Volcanoes	HAZARD EXTENT	LAV	LHR	PDC	BAL	VTS	Remarks
Pitogo	Sulu						
Sinumaan	Sulu						
Tukay	Sulu						
Tumatangas	Sulu						
Kalatungan	Bukidnon						
Labo	Camarines Norte						
Mahagnao	Leyte		✓	✓			
San Cristobal	Laguna, Quezon						
Silay	Negros Occidental						
Cancajanag	Leyte						
Vulcan	Camiguin						
Corregidor	Cavite						
Negron	Pampanga, Zambales						

Ongoing?

To Verify

Not Applicable

Thank You po!